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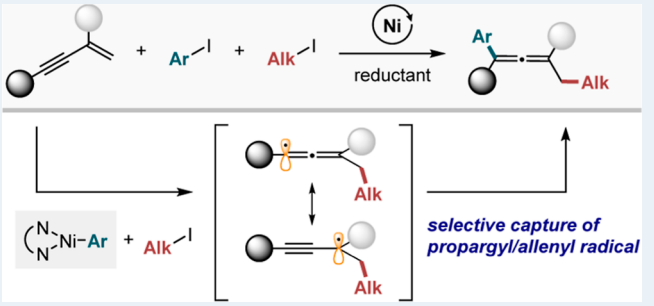
Chemo- and Regioselective Nickel-Catalyzed Reductive 1,4-Alkylarylation of 1,3-Enynes through an L_2NiAr Intermediate

Ji Hwan Jeon,[‖] Gun Ha Kim,[‖] Ho Seung Lee, Da Hye Kim, Soochan Lee, Wonyoung Choe, Byunghyuck Jung,* Jan-Uwe Rohde,* and Sung You Hong*

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ABSTRACT: Three-component reductive dicarbofunctionalization reactions of 1,3-enynes have been rarely reported because of the intricate control of chemo- and regioselectivity required, coupled with a limited understanding of radical and catalytic species involved. Herein, we report a nickel-catalyzed reductive 1,4-alkylarylation method for 1,3-enynes to yield tri- and tetrasubstituted allenes using readily accessible alkyl and aryl iodides, featuring a simple operational protocol and mild reaction conditions. In our mechanistic studies, the formation of a propargyl/allenyl radical was substantiated by the isolation of a propargyl dimer, the detection of the corresponding TEMPO−radical adduct, and radical probe experiments. Two reduced L_2NiAr complexes, expected to act as catalytic intermediates, were generated and characterized by EPR spectroscopy as Ni^I complexes. The stoichiometric reaction of $L_2Ni(p\text{-}NCC_6H_4)$ with 1,3-enyne and alkyl iodide showed conversion into the corresponding propargyl dimer and allene, suggesting that the reaction encompasses the same key mechanistic steps as the catalytic reaction, i.e., activation of alkyl iodide, generation of a propargyl/allenyl radical, and selective coupling of this radical with the aryl component.



KEYWORDS: *allenes, (aryl)nickel(I) complexes, 1,3-enynes, nickel catalysis, radical relay, reductive coupling, three-component reaction*

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1. INTRODUCTION

Allenes are cumulated dienes, which have garnered considerable attention in pharmaceutical science and in the synthetic community because of their distinct biological activity and chemical reactivity.[1] Prototropic and sigmatropic rearrangements, as well as substitution and elimination reactions, represent classic methods for the synthesis of allenes.[1,2] However, these methods often involve laborious steps for preparing activated substrates and require harsh reaction conditions, particularly when aiming to synthesize multisubstituted allenes. In this regard, the catalytic functionalization of readily accessible 1,3-enynes to afford multisubstituted allenes has gained attraction as an efficient and practical synthetic approach.[3] For instance, the Hoveyda, Engle, and Ge groups independently reported CuH-catalyzed hydroboration reactions of 1,3-enynes to give allenyl boronates.[4] Moreover, catalytic hydrosilylation, hydrogenation, hydroalkylation, and hydroarylation reactions of 1,3-enynes using hydride sources have been extensively studied, which likely proceed through metal-hydride complexes generated in situ.[5]

Unlike hydrofunctionalization, dicarbofunctionalization of 1,3-enynes, which directly forms two C−C bonds, has emerged as a powerful synthetic approach for the efficient synthesis of allenes. However, the polar approach via allenyl anion species typically involves the use of organolithium reagents, which often limits the tolerance for functional groups and necessitates cryogenic cooling (Scheme 1A).[6] Recently, there has been remarkable progress in radical-involved dicarbofunctionalization of unsaturated systems, including 1,3-enynes, by merging single-electron transfer processes and transition-metal catalysis.[7,8] Radical-based redox-neutral 1,4-dicarbofunctionalization of 1,3-enynes has been demonstrated (Scheme 1B). These methods offer the advantage of efficient twofold C−C bond formation.

Cross-electrophile coupling (XEC) reactions of π systems, utilizing readily available organic (pseudo)halides, have recently garnered significant attention in the synthetic community. This is attributed to their mild reaction conditions, simple operations, and improved functional group compatibility.[9] The application of the XEC approach to three-component reductive difunctionalization has been demonstrated for alkenes as the conjunctive platform.[10] High levels of chemo- and regioselectivity were achieved by utilizing activated alkenes to enable the philicity-

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Scheme 1. Overview of Catalytic Dicarbofunctionalization of 1,3-Enynes and Our Approach

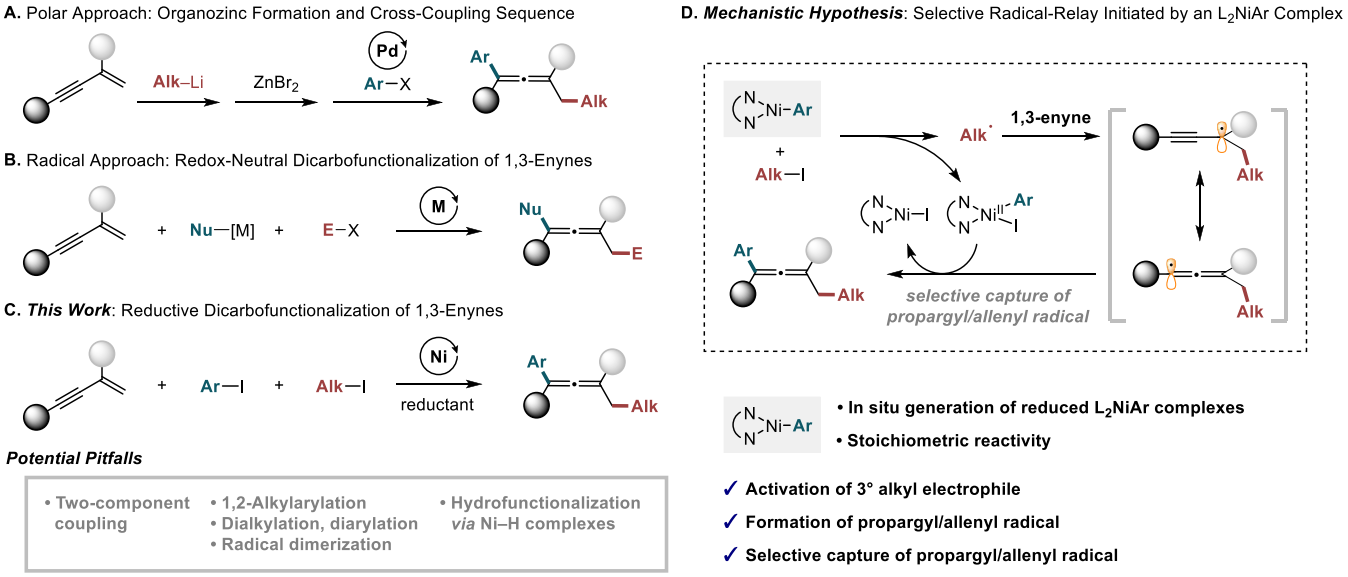
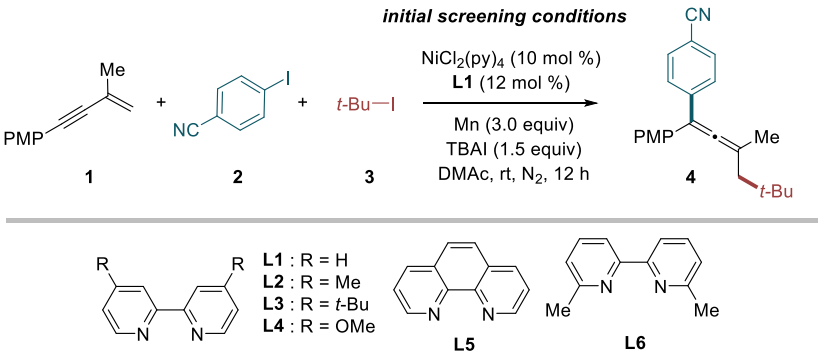


Table 1. Optimization of Reaction Conditions[a]



entry	change from above conditions	yield (%)
1	none	88 (88)[b]
2	$NiCl_2$·glyme instead of $NiCl_2(py)_4$	57
3	$Ni(OTf)_2$ instead of $NiCl_2(py)_4$	78
4	**L2** instead of **L1**	86
5	**L3** instead of **L1**	84
6	**L4** instead of **L1**	61
7	**L5** instead of **L1**	80
8	**L6** instead of **L1**	12
9	Zn instead of Mn	47
10	TDAE instead of Mn	5
11	without TBAI	40
12	TMSCl instead of TBAI	41
13	**Mn (1.5 equiv), TBAI (0.75 equiv), 1 h**	**89 (88)**[b]

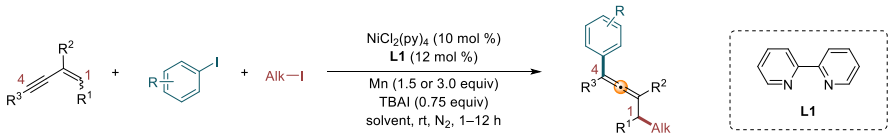
[a]Initial screening conditions: 1,3-enyne **1** (0.30 mmol, 1.5 equiv), aryl iodide **2** (0.20 mmol, 1.0 equiv), alkyl iodide **3** (0.30 mmol, 1.5 equiv), $NiCl_2(py)_4$ (10 mol %), **L1** (12 mol %), Mn (3.0 equiv), TBAI (1.5 equiv), DMAc (1 mL), rt, 12 h. Yields were determined by GC using *n*-dodecane as an internal standard. DMAc = *N*,*N*-dimethylacetamide; glyme = ethylene glycol dimethyl ether; OTf = trifluoromethanesulfonate; PMP = *para*-methoxyphenyl; TBAI = tetra-*n*-butylammonium iodide; TDAE = tetrakis(dimethylamino)ethylene; and TMSCl = trimethylsilyl chloride. [b]Isolated yield is given in parentheses.

matched addition of C-centered radicals or by employing alkenes bearing preinstalled directing groups.[11,12]

Three-component XEC reactions utilizing 1,3-enynes, as opposed to alkenes, are substantially more challenging. The presence of competing C(sp^2) and C(sp) sites in 1,3-enynes demands precise control of chemo- and regioselectivity for the incorporation of two distinct electrophiles. To facilitate the reductive three-component 1,4-dicarbofunctionalization of the 1,3-enyne substrate (Scheme 1C) in competition with vicinal difunctionalization, dialkylation, diarylation, and recombinative radical dimerization, it is essential to achieve the selective capture of a propargyl/allenyl radical at the allenyl C(sp) site, rather than capture at the propargyl C(sp^2) site or capture of an alkyl radical, by the active Ni catalyst.[13] Furthermore, undesired

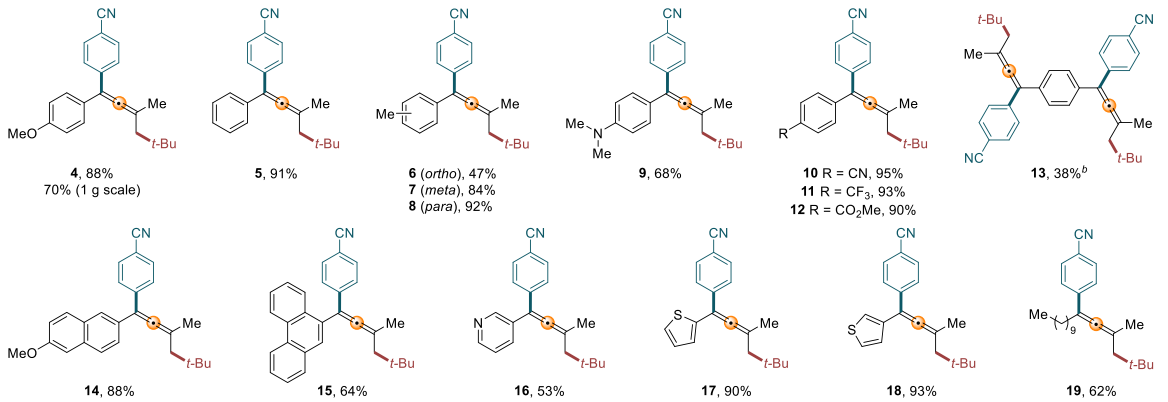
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Table 2. Reaction Scope[a]



Scope of 1,3-Enynes[a]

R^1 = H, R^2 = Me, R^3 = Aryl or Alkyl



4, 88%
70% (1 g scale)

5, 91%

6 (*ortho*), 47%
7 (*meta*), 84%
8 (*para*), 92%

9, 68%

10 R = CN, 95%
11 R = CF_3, 93%
12 R = CO_2Me, 90%

13, 38%[b]

14, 88%

15, 64%

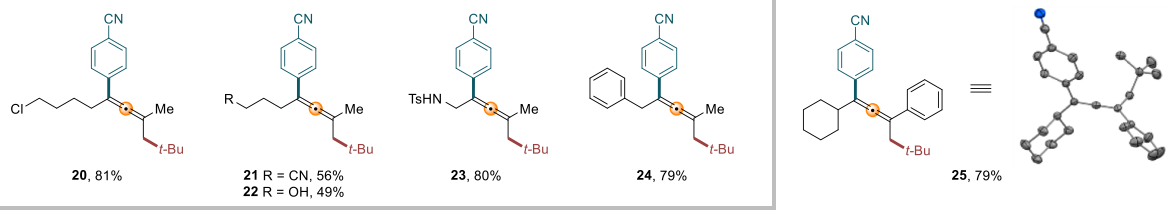
16, 53%

17, 90%

18, 93%

19, 62%

R^1 and R^2 Substituents[a]



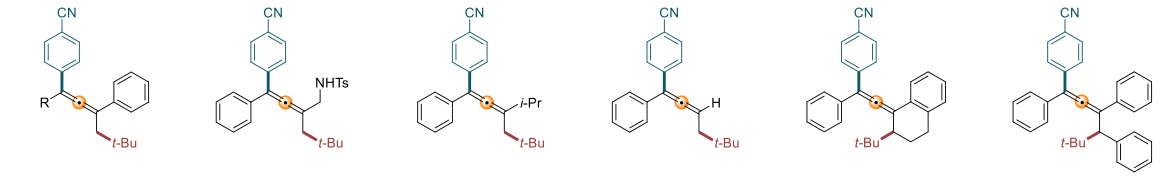
20, 81%

21 R = CN, 56%
22 R = OH, 49%

23, 80%

24, 79%

25, 79%



26 R = *t*-Bu, 44%
27 R = Ph, 58%

28, 48%

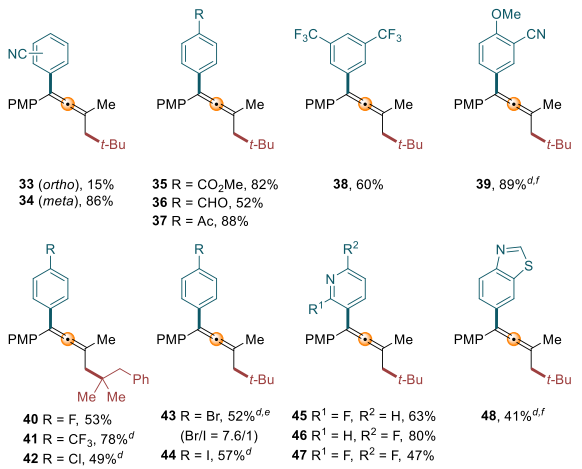
29, 57%

30, 39%

31, 47%, dr[c] > 99:1

32, 33%, dr[c] = 2.8:1

Scope of Aryl Iodides[a]



33 (*ortho*), 15%
34 (*meta*), 86%

35 R = CO_2Me, 82%
36 R = CHO, 52%
37 R = Ac, 88%

38, 60%

39, 89%[d,f]

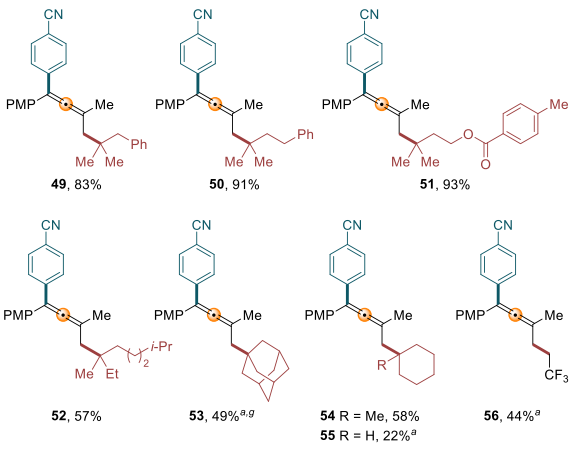
40 R = F, 53%
41 R = CF_3, 78%[d]
42 R = Cl, 49%[d]

43 R = Br, 52%[d,e]
(Br/I = 7.6/1)
44 R = I, 57%[d]

45 R^1 = F, R^2 = H, 63%
46 R^1 = H, R^2 = F, 80%
47 R^1 = F, R^2 = F, 47%

48, 41%[d,f]

Scope of Alkyl Iodides[d]



49, 83%

50, 91%

51, 93%

52, 57%

53, 49%[a,g]

54 R = Me, 58%
55 R = H, 22%[a]

56, 44%[a]

[a]Reaction conditions: 1,3-enyne (0.60 mmol), aryl iodide (0.40 mmol), alkyl iodide (0.60 mmol), $NiCl_2(py)_4$ (10 mol %), **L1** (12 mol %), Mn (1.5 equiv), TBAI (0.75 equiv), DMAc (2 mL), rt. Isolated yields. [b]1,3-enyne (0.40 mmol), aryl iodide (1.2 mmol), and alkyl iodide (1.2 mmol) were used. [c]The dr values were determined by ^{1}H NMR spectroscopy. [d]Reaction conditions: 1,3-enyne (0.60 mmol), aryl iodide (0.40 mmol), alkyl iodide (1.2 mmol), $NiCl_2(py)_4$ (10 mol %), **L1** (12 mol %), Mn (3.0 equiv), TBAI (0.75 equiv), THF (2 mL), rt, reaction carried out in the dark. [e]The Br/I ratio was determined by ^{1}H NMR spectroscopy. [f]DMAc instead of THF. [g]Alkyl iodide (1.8 mmol) and Mn (3.0 equiv) were used.

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hydrofunctionalization may prevail when alkyl halides are present in Ni-catalyzed reactions under reductive conditions, which may generate NiH species through the oxidative addition of alkyl halide and subsequent β-hydrogen elimination.[14] Recent examples of XEC reactions of 1,3-enynes utilized photochemistry or electrochemistry for electrophile activation, in addition to Ni catalysis.[8a,15] The reactivity of different sites of the 1,3-enyne substrate was examined through DFT calculations.[15b] However, despite the progress in the development of reductive dicarbofunctionalization of unsaturated systems, the mechanistic understanding of the reactivity of radical intermediates and catalytically active species remains limited.

Herein, we report an operationally straightforward Ni-catalyzed reductive 1,4-alkylarylation of conjugated 1,3-enynes using a metal reducing agent as a complementary synthetic route to furnish tri- or tetrasubstituted allenes (Scheme 1C). In studies on related reductive Ni catalysis with unsaturated substrates,[12c,13b] reduced Ni complexes such as L_2NiX and L_2NiAr were proposed to initiate the reaction through the activation of alkyl halide to generate an alkyl radical. In the three-component XEC reaction described here, an alkyl radical generated in situ may react with the 1,3-enyne to give a propargyl/allenyl radical (Scheme 1D). We present mechanistic evidence of the presence and involvement of a propargyl/allenyl radical as an intermediate in this reaction. The excellent chemo- and regioselectivity observed in the allene product formation suggests selective coupling of this radical with the aryl component. Furthermore, we report the generation, characterization, and reactivity of reduced L_2NiAr complexes (Scheme 1D), demonstrating that they can function as catalytic intermediates in XEC reactions of 1,3-enynes.

2. RESULTS AND DISCUSSION

2.1. Reaction Optimization. Catalytic 1,4-dicarbofunctionalization of 1,3-enyne **1** was conducted using *para*-iodobenzonitrile (**2**) and *tert*-butyl iodide (**3**) as model aryl and alkyl electrophiles. After extensive screening of reaction parameters (see also the Supporting Information, Section 3), the optimized reaction conditions resulted in the formation of tetrasubstituted allene **4** as a single regioisomer in 88% yield (Table 1, entry 1). The combination of $NiCl_2(py)_4$ and 2,2′-bipyridine (**L1**) turned out to be the optimal precatalyst (entries 2−8). It is of note that the 1,2-difunctionalized regioisomer was also not observed with ligands **L2**−**L6**. Substituting Zn or TDAE for Mn as the reducing agent resulted in decreased yields (entries 9 and 10). The absence of TBAI significantly diminished the yield of the product (entry 11). Presumably, TBAI facilitates the electron transfer from Mn by improving homogeneity.[16] The use of TMSCl did not improve the yield (entry 12).[17] The reaction conditions were further optimized by reducing the amount of Mn to 1.5 equiv and that of TBAI to 0.75 equiv, while shortening the reaction time to 1 h (entry 13 and Supporting Information, Section 3).

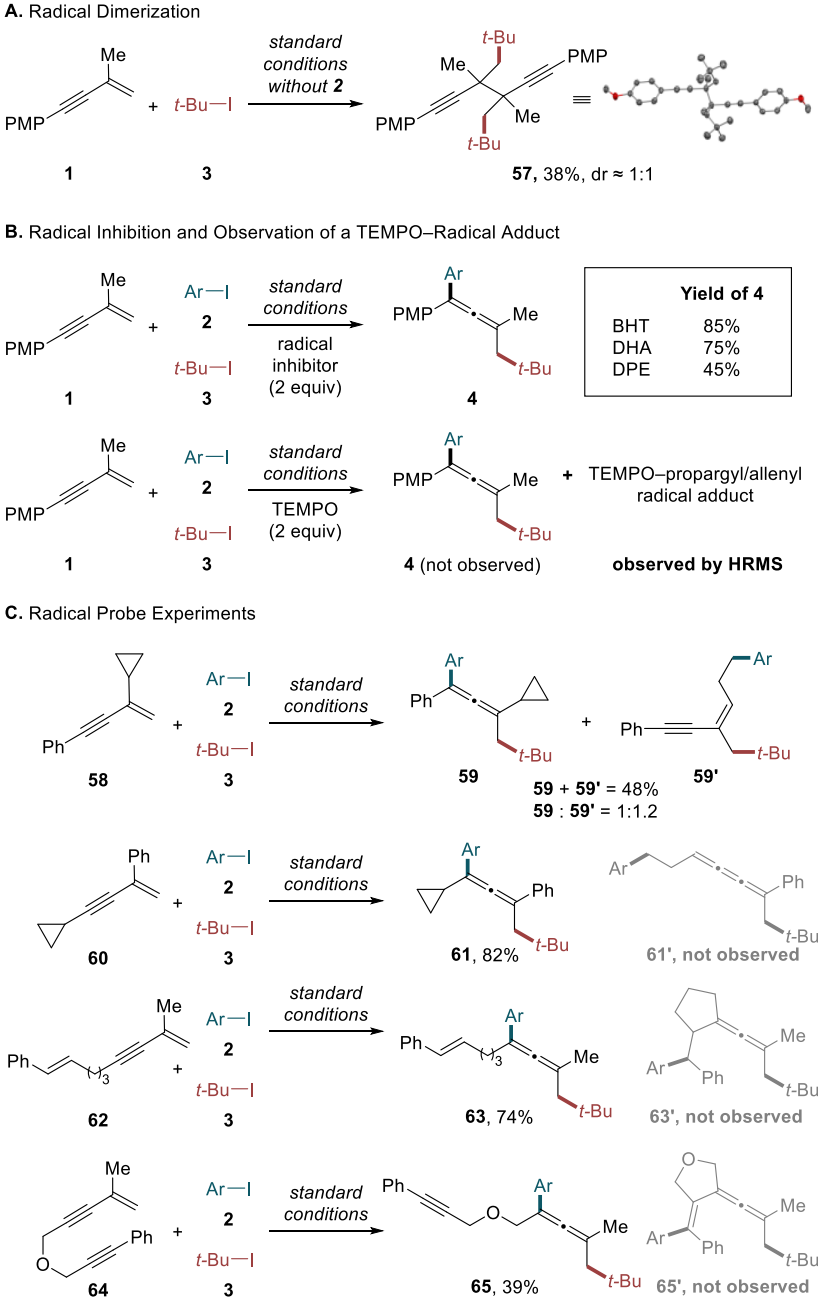
2.2. Reaction Scope. With the optimized conditions, we examined 1,3-enynes with a diverse range of substituents (R^1, R^2, and R^3), as depicted in Table 2. Various aryl substituents in R^3 position exhibited excellent tolerance toward the reductive 1,4-alkylarylation reaction affording the tetrasubstituted allene products. Good to excellent yields (68−95%) were observed regardless of the electronic nature of the aryl substituents of **4**, **5**, and **7**−**12**, which indicates the involvement of a radical intermediate rather than anionic or cationic intermediates. However, the conversion of a 1,3-enyne with an *ortho*-methyl substituent in the R^3 arene ring required an extended reaction time to yield allene **6** in 47% yield, presumably because of steric hindrance. Interestingly, symmetric bis(allene) **13** was also obtained through double 1,4-alkylarylation in 38% yield.[8g] Fused arenes were also incorporated giving **14** and **15** in 88 and 64% yield, respectively. The sterically encumbering 9-phenanthrenyl substituent showed slightly diminished efficiency. Among heteroaromatic substituents, the 3-pyridyl moiety provided product **16** in 53% yield. The 2- or 3-thienyl group afforded the corresponding allenes **17** and **18** in yields of 90 and 93%, respectively. To our delight, the desired allene products **19**−**21** and **24** were readily obtained in 56−81% yield from 1,3-enynes with alkyl, chloroalkyl, cyanoalkyl, and benzyl substituents in the position of R^3. 1,3-Enynes with acidic hydroxy and amide groups were well tolerated to give the allenes **22** and **23** in yields of 49 and 80%, respectively.

Reactions of 1,3-enynes with various R^2 substituents furnished the corresponding allene products in 33−79% yield. Substrates with a phenyl substituent in R^2 underwent the desired 1,4-alkylarylation reaction to generate the allenes **25**−**27** in 44−79% yield. Reactions of 1,3-enynes with alkyl substituents in R^2 position delivered the corresponding products in moderate yields (48 and 57% for **28** and **29**, respectively). The formation of **23** and **28** highlights the synthetic utility of our method as a rapid approach to allene precursors of heterocyclic compounds by intramolecular cyclization.[1b,8e] The reaction of a monosubstituted 1,3-enyne (**S30**) gave the desired allene in a diminished yield (39% for **30**) along with the 1,2-difunctionalization product (7%). The decreased regioselectivity in this reaction suggests that it depends on the steric effect of the R^2 substituent (Supporting Information, Section 4). Furthermore, the dicarbofunctionalization of trisubstituted 1,3-enynes was examined. The reactions using a fused-cyclic 1,3-enyne and a triphenyl-substituted 1,3-enyne yielded allene products **31** and **32** in yields of 47 and 33%, respectively. Notably, allene **31**, having an exocyclic double bond, was obtained as a single diastereomer, but a reduced diastereomeric ratio was observed for unfused allene **32**.

Next, we evaluated the aryl iodide scope in this protocol. When *para*-iodobenzonitrile was replaced with *meta*-iodobenzonitrile, the reaction afforded allene **34** in a yield similar to that of **4** (86 vs 88%). However, when *ortho*-iodobenzonitrile was used, the yield of the allene (**33**) was drastically reduced to 15%, indicating a detrimental steric effect of the *ortho* substituent. Ester, formyl, and acetyl moieties were compatible with the reaction conditions (**35**−**37**, 52−88% yield). We also investigated F- and CF_3-substituted aryl iodides, which find extensive application in pharmaceutical science and materials chemistry. These substrates successfully underwent the reactions leading to the formation of allenes **38, 40, 41**, and **45**−**47** in 47−80% yield. When we tried to use electron-neutral or electron-rich aryl iodides in this method, the reactions resulted in complex mixtures and the desired allene products were not observed (Figure S1). An aryl iodide substituted with both an electron-donating methoxy group and an electron-withdrawing cyano group was converted into allene **39** in 89% yield. Although electron-deficient aryl iodides showed better yields, the formation of **39** from this aryl iodide implies that the reaction is not limited to aryl iodides bearing only electron-withdrawing groups. Halide compatibility was tested using 1-halo-4-iodobenzenes (F-, Cl-, Br-, and I-substituted aryl iodides). 1-Chloro-4-iodobenzene gave allene **42** in 49% yield. 1-Bromo-4-iodobenzene showed good Br/I selectivity to give

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Scheme 2. Control, Inhibition, and Radical Probe Experiments[a]



[a]Ar = *p*-NCC_6H_4.

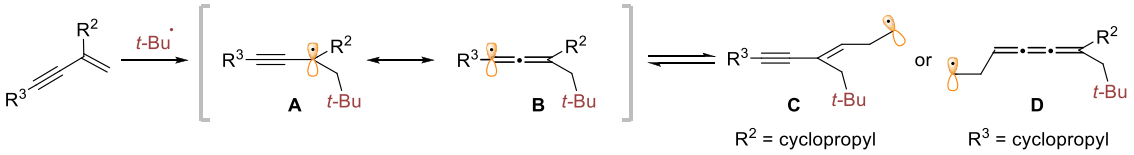
allene **43** as the major product (7.6:1 ratio) in a mixture with iodophenyl-substituted allene **44**. The use of 1,4-diiodobenzene as aryl electrophile resulted in the formation of allene **44** in 57% yield without any sign of formation of the corresponding bis(allene). Heteroaromatic electrophiles furnished the products in moderate to good yields (**45**–**48**, 41–80%). The formation of allene **48** from 6-iodobenzo[*d*]thiazole also demonstrated the compatibility of an electron-rich heteroaryl iodide.

The reaction was amenable to *tert*-alkyl iodides having acyclic and cyclic alkyl groups. Acyclic alkyl iodides provided the products in good to excellent yields (**49**–**52**, 57–93%). The 1-adamantyl and 1-methylcyclohexyl groups were successfully incorporated to afford allene products **53** and **54** in yields of 49 and 58%, respectively. Secondary and primary alkyl iodides also gave the allene products (**55** and **56**), albeit in diminished yields.

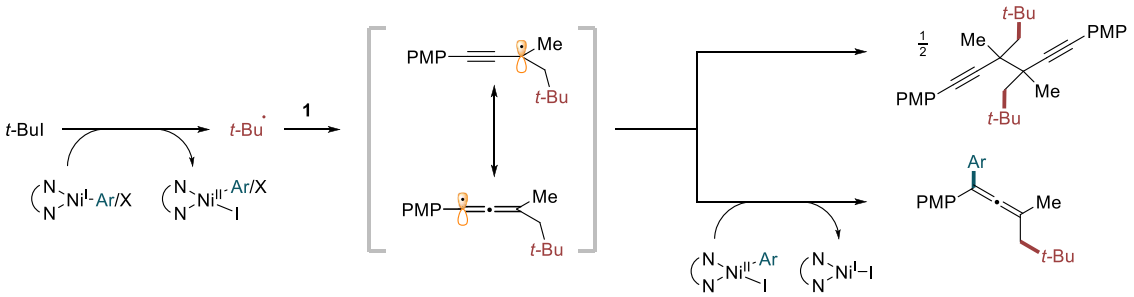
2.3. Evidence of a Propargyl/Allenyl Radical Intermediate. Upon activation of alkyl iodide by the active Ni catalyst, the addition of an alkyl radical to a 1,3-enyne may take place at multiple sites.[15b] Thus, depending on the position of the radical addition, a diverse array of radicals can be formed. To confirm the presence of a radical intermediate derived from the 1,3-enyne substrate and to elucidate its identity, we carried out several experiments. The reaction of 1,3-enyne **1** and alkyl iodide **3** in the absence of aryl iodide afforded product **57**, which was isolated and accounted for 38% conversion of **1** (Scheme 2A). Its structure was determined by X-ray crystallography. The formation of **57** is explained by the recombination of two

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Scheme 3. Resonance Structures of Radical Species and Ring-Opening



Scheme 4. Proposed Mechanism of Formation of the Propargyl Dimer and Allene



propargyl radicals. The propensity of a propargyl radical to exist in resonance with its allenyl counterpart, as depicted in Scheme 3, radical species **A** and **B**, aligns with the observed chemo- and regioselectivity in the formation of allene **4**.

The observation of the propargyl dimer **57** prompted us to probe the propargyl/allenyl radical further. While butylated hydroxytoluene (BHT) had little effect on the 1,4-alkylarylation of **1** (yield of **4**, 85%), 9,10-dihydroanthracene (DHA) and 1,1-diphenylethylene (DPE) diminished the yield of the allene product (75 and 45% for DHA and DPE, respectively). In the presence of (2,2,6,6-tetramethylpiperidin-1-yl)oxyl (TEMPO, 2 equiv with respect to **2**, Scheme 2B), the 1,4-alkylarylation of **1** was completely inhibited and most of the enyne and the aryl iodide remained unreacted.[8] Presumably, TEMPO deactivated the Ni catalyst, as it is known to form Ni(TEMPO)$_2$.[18] The high-resolution mass spectrum of a sample of the reaction mixture exhibited a peak having the mass and isotope distribution pattern expected for the TEMPO adduct of the propargyl/allenyl radical (m/z = 386.3057, Figure S2), indicating that at least a small fraction of **1** still reacted to form the radical intermediate.

To elucidate the involvement of a free propargyl/allenyl radical in the catalytic reaction, we introduced radical probe substituents in the R^2 or R^3 position of the 1,3-enyne (Scheme 3). The presence of a cyclopropyl substituent, for example, on either the propargyl radical C(sp^2) center (R^2) or the allenyl radical C(sp) center (R^3) may allow isomerization of the radical intermediate to a ring-opened radical species (Scheme 3, radical species **C** for R^2 = cyclopropyl or radical species **D** for R^3 = cyclopropyl). Such isomerization depends on the product-forming rate of consumption of the propargyl/allenyl radical by the active Ni catalyst. Under standard reaction conditions, the use of a 1,3-enyne with a cyclopropyl group installed in the position of R^2 (**58**, Scheme 2C) afforded allene **59** and the ring-opened enyne **59′**.[8b,c,g] These products were obtained as an inseparable mixture (48% total yield) with a ratio of 1:1.2, as determined by ^{1}H NMR spectroscopy.

These results suggest that the propargyl/allenyl radical is an intermediate in the catalytic reaction and that it undergoes coupling with the aryl component either at the allenyl radical center (**B**) to give **59** or at the ring-opened alkyl radical center (**C**) to give **59′**. No product was observed from coupling at the propargyl radical center (**A**), as was the case for other 1,3-enynes examined (Table 2). In contrast, standard reactions using a 1,3-enyne with a ring-closing or ring-opening radical probe installed in R^3 position (**60**, **62**, and **64**, Scheme 2C) yielded only the nonisomerized products **61**, **63**, and **65** in yields of 82, 74, and 39%, respectively.[8d−g] Surprisingly, although enyne **60** bears the same radical probe (cyclopropyl group) as enyne **58**, the isomerized product **61′**, which would have been derived from **D** (Scheme 3), was not observed at all. These observations indicate that the consumption rate of the propargyl/allenyl radical derived from **58** (R^2 = cyclopropyl) is comparable with the rate of ring-opening and coupling of **C** but that the consumption rate of the propargyl/allenyl radical derived from **60** (R^3 = cyclopropyl) is faster than the rate of ring-opening and coupling of **D**.

2.4. Generation, Characterization, and Stoichiometric Reactions of Reduced L$_2$NiAr Complexes. The isolation of the propargyl dimer (**57**), the detection of a TEMPO−propargyl/allenyl radical adduct by high-resolution mass spectrometry, and the radical probe experiments together indicate the intermediacy of a propargyl/allenyl radical (Scheme 4). The formation of this radical can be explained by the reaction of **1** with the *tert*-butyl radical, which may be generated from the reaction of **3** with a reduced Ni complex, such as L$_2$NiAr or L$_2$NiX. The initiation step is supported by literature on the reactions of other reduced Ni complexes with alkyl iodides.[12c,13] To account for the product formation observed in our substrate scope studies, we hypothesize that the propargyl/allenyl radical is captured by a Ni^{II} complex carrying the aryl component, L$_2$Ni^{II}ArI (Scheme 4), followed by reductive elimination of the allene. Similar radical capture events by L$_2$Ni^{II}ArX complexes were previously described.[13b,19]

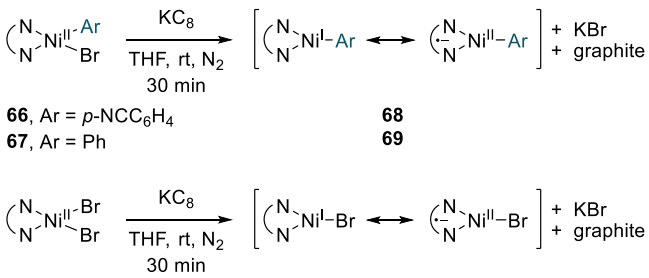
Based on the mechanistic working hypothesis (Scheme 4), we pursued the generation of reduced L$_2$NiAr complexes to test whether these complexes can act as intermediates and mediate allene product formation in reaction with **1** and **3**. Because the catalytic reaction conditions were compatible with electron-deficient aryl iodides, such as 4-cyanophenyl iodide (Table 2), but not with more electron-donating aryl iodides, such as phenyl iodide (Table S8), we decided to investigate the two L$_2$NiAr complexes with Ar = *p*-NCC$_6$H$_4$ and Ph.

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To access these complexes, we selected the corresponding L_2NiArX complexes as suitable precursors. From among the *N*,*N*′-bidentate supporting ligands that showed good catalytic activity (Table 1, entries 1 and 4−7, **L1**−**L5**), **L5** was selected, as the preparation of related Ni^{II}(**L5**)(Ar)Br complexes has been well described. The complexes Ni^{II}(**L5**)(*p*-NCC_6H_4)Br (**66**) and Ni^{II}(**L5**)(Ph)Br (**67**) were readily prepared by oxidative addition of 4-cyanophenyl bromide and phenyl bromide, respectively, to Ni^0(**L5**)(cod), using an adapted literature procedure.[13b,20] Complexes **66** and **67** were isolated in good yields (86 and 88%, respectively) and were pure by elemental analysis (Supporting Information, Section 5.2.1).

We next established reaction conditions for the reduction of **66** and **67**. While we employed Mn or Zn as reducing agents in the catalytic reactions, these metals were not suitable for stoichiometric reactions. Oxidation of Mn gives the Mn^{2+} ion (S = 5/2), whose intense EPR signal would interfere with the characterization of reduced Ni(**L5**)(Ar) (S = 1/2) complexes by EPR spectroscopy. The use of Zn would require a prolonged reaction time or an excess amount of it.[13] Therefore, we decided to use KC_8 as an alternative reducing agent.[21] Complexes **66** and **67** were reduced with 1 equiv of KC_8 in THF (t = 30 min, Scheme 5).

Scheme 5. Generation of Reduced Ni(L5)(Ar) Complexes and Ni(L5)Br



Upon addition of KC_8 to a suspension of **66** in THF, its orange-red color turned dark brown. The EPR spectrum of the reaction mixture measured at 100 K (Figures 1 and S3) shows two signals, that is, a rhombic signal taking up most of the signal intensity (g = [2.265, 2.128, 2.053]) and a fairly sharp signal of very low intensity (g = 2.003, peak-to-peak separation of 13 G). The rhombic signal accounts for 31% of the Ni present and the sharp signal for less than 1% (Supporting Information, Section 5.2.3). The target complex Ni(**L5**)(*p*-NCC_6H_4) (**68**) can be described by the limiting spin configurations of a Ni^I complex and a Ni^{II}-ligand-radical complex, as shown in Scheme 5. The spectral parameters of the rhombic signal suggest that the unpaired electron spin density largely resides in a metal-based orbital. Similar EPR parameters were reported for other L_2Ni^IAr and L_2Ni^IX complexes with *N*,*N*′-bidentate ligands in DMAc, Et_2O, THF, 2-Me-THF, or toluene (9−77 K).[13b,22,23b,24a] In contrast, the related complex Ni(**L5**)(Mes) was described as a ligand-radical complex,[13b,25] giving rise to an isotropic signal at g = 2.009 (DMAc, 10 K), when obtained from reduction of Ni(**L5**)(Mes)Br with Zn, where Mes = 2,4,6-trimethylphenyl.[13b] Thus, based on the major signal observed in the EPR spectrum of **68**, this complex is best described as a Ni^I complex of the unreduced phen ligand, Ni^I(**L5**)(*p*-NCC_6H_4). The minor signal at g = 2.003 may be attributed to an organic radical or perhaps a ligand-radical complex of Ni^{II},[26] which could be a solvent-stabilized analogue of **68** such as Ni^{II}(**L5**$^{\bullet-}$)(*p*-NCC_6H_4)(thf).[22]

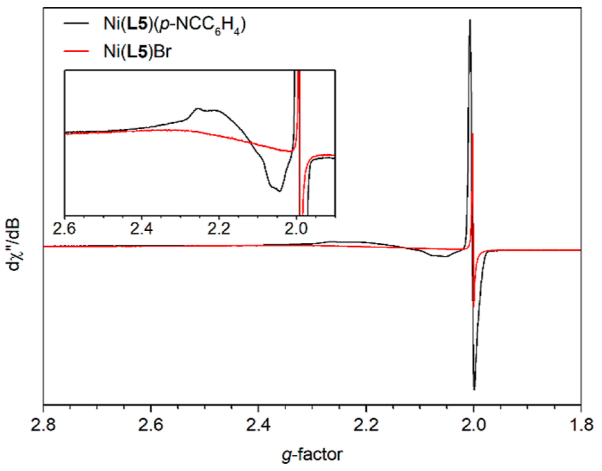


Figure 1. EPR spectra (X-band, 100 K) of Ni(**L5**)(*p*-NCC_6H_4) (**68**, black line) and Ni(**L5**)Br (red line) generated in situ in THF. Inset: expanded view.

For comparison, Ni(**L5**)Br was generated from Ni^{II}(**L5**)Br_2 using the same method as the one employed for the reduction of **66**. The EPR spectrum of the reaction mixture (100 K, Figures 1 and S4) shows a broad signal (g = 2.126), consistent with Ni^I(**L5**)Br, along with a narrow signal (g = 2.002, peak-to-peak separation of 4 G). Although this spectrum appears to be similar to that reported for this complex in DMAc (10 K),[13b] at first glance, the spectrum acquired in this study clearly shows two distinct features (Figure S4). These differences could be due to changes in the solvent or the measurement temperature. Importantly, because the EPR spectra of **68** and Ni(**L5**)Br reported here were acquired under the same conditions on compounds prepared using the same method, the different signals confirm that the reduction product of **66** is Ni(**L5**)(*p*-NCC_6H_4) and not Ni(**L5**)Br. This is noteworthy because ligand exchange between L_2NiAr and unreacted L_2NiArBr may alternatively lead to the formation of L_2NiBr and L_2NiAr_2.[23,24]

The consumption of **66** and formation of **68** were verified by UV−vis spectroscopy and mass spectrometry. The UV−vis spectrum of a solution of **66** in THF shows a broad band with an absorption maximum at 454 nm (Figure 2). This spectrum is similar to that reported for Ni^{II}(**L5**)(*o*-tolyl)Br in THF (λ_{max}, 487 nm).[13b] The spectrum of the diluted and filtered reaction mixture of **68** (Supporting Information, Section 5.2) shows a new band at 376 nm and a shoulder at 475 nm (Figure 2) and does not contain the broad band of **66**. The high-resolution mass spectrum of the reaction mixture shows a peak having the mass and isotope distribution pattern expected for {**68** + H}$^+$ (m/z = 341.0456, Figure S10). Taken together, the reaction produced **68** as the only identifiable Ni complex in solution.

Complex **67** was reduced in the same manner and the reaction mixture was analyzed by EPR and UV−vis spectroscopy. Upon addition of KC_8, the maroon suspension of **67** in THF turned dark orange-brown. The EPR spectrum of the reaction mixture (100 K, Figure S5) shows a broad signal with fine structure (g = 2.220) and a narrow signal of very low intensity (g = 2.002, peak-to-peak separation of 4 G). The broad signal can be assigned to Ni^I(**L5**)(Ph) (**69**) and accounts for 71% of the Ni present, while the narrow signal represents less than 1% (Supporting Information, Section 5.2.3).

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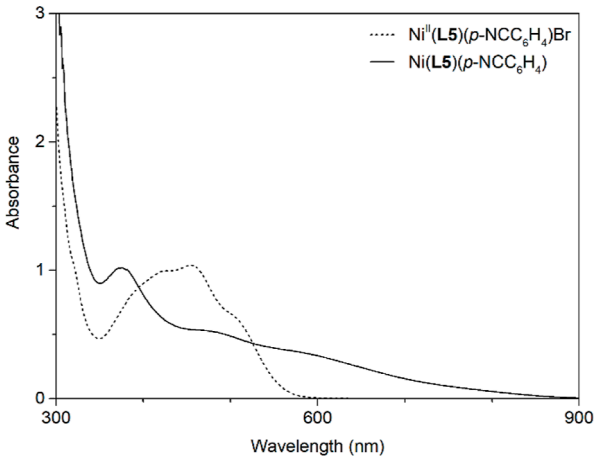


Figure 2. UV−vis spectra of 0.5 mM Ni^{II}(**L5**)(*p*-NCC_6H_4)Br in THF (**66**, dotted black line; path length, 0.5 cm) and of Ni(**L5**)(*p*-NCC_6H_4) generated in situ in THF (**68**, solid black line; path length, 0.5 cm). After the generation of **68**, the suspension was diluted to approximately 0.5 mM Ni and then filtered.

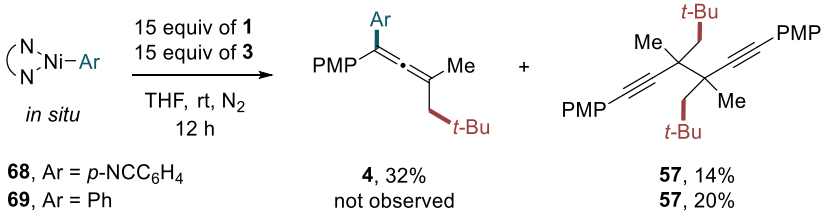
The UV−vis spectra of **67** in THF and the reaction mixture confirmed the conversion of **67** into **69** (Figure S6). Consistent with the structural similarities expected for pairs of complexes differing only in the *para* position of the aryl ligand, the spectral features of **67** (λ_{max}, 480 nm) and **69** (λ_{max}, 378 and 477 nm) are very similar to those of **66** and **68**, respectively (Figures S7 and S8). The complexes Ni^{II}(**L5**)Br_2 and Ni^{I}(**L5**)Br are only poorly soluble in THF, thus limiting comparison by UV−vis spectroscopy (Figure S9).

Having established conditions for the accumulation of **68** and **69**, we explored the stoichiometric reaction of these complexes with 1,3-enyne **1** and alkyl iodide **3** (Scheme 6).[27] To reflect the molar ratio of reactants used under standard conditions [0.040 mmol of $NiCl_2(py)_4$, 0.60 mmol of **1**, and 0.60 mmol of **3**], the amounts of each **1** and **3** were set to 15 equiv with respect to Ni. The substrates **1** and **3** were added 30 min after the addition of KC_8 to **66** or **67**. The reaction of **68** with **1** and **3** afforded two products, allene **4** (32% wrt Ni) and the radical dimer **57** (14%, Scheme 6). These are the same products as the ones observed under catalytic conditions (Table 1, entries 1 and 7) and in the absence of aryl iodide (Scheme 2A), respectively. Therefore, Ni(**L5**)(*p*-NCC_6H_4) (**68**) is a possible intermediate capable of activating alkyl iodide and delivering the aryl component to produce the allene. Because the yield of allene **4** is similar to the accumulation of mononuclear **68** prior to the stoichiometric reaction (31% wrt Ni, as detected by EPR spectroscopy), the conversion of **68** may be quantitative. However, a portion of the remaining 69% of Ni may be present as a dinuclear Ni^{I} complex (S = 1 or 0),[23b,24a,28] which could be in equilibrium with **68** and thus produce more active Ni^{I} complex in the course of the reaction. The formation of radical dimer **57** requires additional Ni^{I} for activation of alkyl iodide and generation of propargyl/allenyl radical. This may be achieved by Ni(**L5**)I, which is expected to be produced in the coupling step of the reaction (Scheme 4), or by an additional amount of **68** released from an undetected dinuclear Ni^{I} complex, if present.

In contrast, the reaction of **69** with **1** and **3** did not deliver the coupled allene product but only the radical dimer **57** (20% wrt Ni, Scheme 6). This result shows that **69** can promote the formation of the propargyl/allenyl radical but not its coupling with the Ph group, suggesting that radical capture or reductive elimination or both are not favorable in this case despite the higher accumulation of mononuclear **69** (71%, as detected by EPR spectroscopy). The contrast in the reactivity of **68** and **69** is consistent with our results observed for the catalytic reaction, which operates with the electron-deficient aryl iodide **2** (Tables 1 and 2) but not with the more electron-rich phenyl iodide (Table S8). In the latter case, only compounds derived from the propargyl/allenyl radical by dimerization and by reaction with *tert*-butyl radical were observed by GC−MS (Figure S1). Similarly, in a related study involving benzyl radical capture by $L_2Ni^{II}ArX$ complexes (using bidentate N-donor ligands), complexes with electron-deficient aryl ligands afforded the coupled Ar−Bn product in high yield.[19]

2.5. Proposed Mechanism. A proposed catalytic cycle for the reductive 1,4-alkylarylation of 1,3-enynes is outlined in Scheme 7. The active catalyst $L_2Ni^{I}I$ (**I**) is generated by the reduction of the NiX_2/L_2 precatalyst in the presence of TBAI. Oxidative addition of aryl iodide to **I** gives $L_2Ni^{III}ArI_2$ (**II**),[24a] which in turn may be reduced by Mn to give $L_2Ni^{II}ArI$ (**III**). (Two alternative pathways for the formation of **III** by comproportionation or through a Ni^0 complex are described in the Supporting Information, Section 5.2.7.[24b]) Further reduction of **III** may afford $L_2Ni^{I}Ar$ (**IV**). Activation of alkyl iodide to give alkyl radical **V** can be achieved by reduced Ni complex **I** or **IV**. The addition of **V** to the 1,3-enyne generates the propargyl/allenyl radical (**VI** and **VII**). The presence of the propargyl/allenyl radical was supported by the detection of its TEMPO adduct by mass spectrometry, the isolation of dimer **57**, and the formation of isomerized product **59′** obtained from the reaction of radical-probe-substituted 1,3-enyne **58** (Scheme 2). Capture of this radical by Ni^{II} complex **III** forms L_2Ni^{III}(aryl)(allenyl)I (**VIII**),[29] from which reductive elimination gives the allene and regenerates Ni catalyst **I**. We speculate that the regioselective formation of the 1,4-difunctionalized allene, as opposed to the 1,2-difunctionalized product, may be due to the steric effect of the R^2 substituent of the 1,3-enyne.[30] This mechanism agrees with the results of the stoichiometric reaction of Ni(**L5**)(*p*-NCC_6H_4) (**68**) with alkyl iodide **3** and 1,3-enyne **1**, which afforded the 1,4-alkylarylation product of **1** (Scheme 6), demonstrating that reduced Ni complex **IV** can mediate allene formation. Furthermore, the

Scheme 6. Stoichiometric Reactions of Reduced Ni(L5)(Ar) Complexes



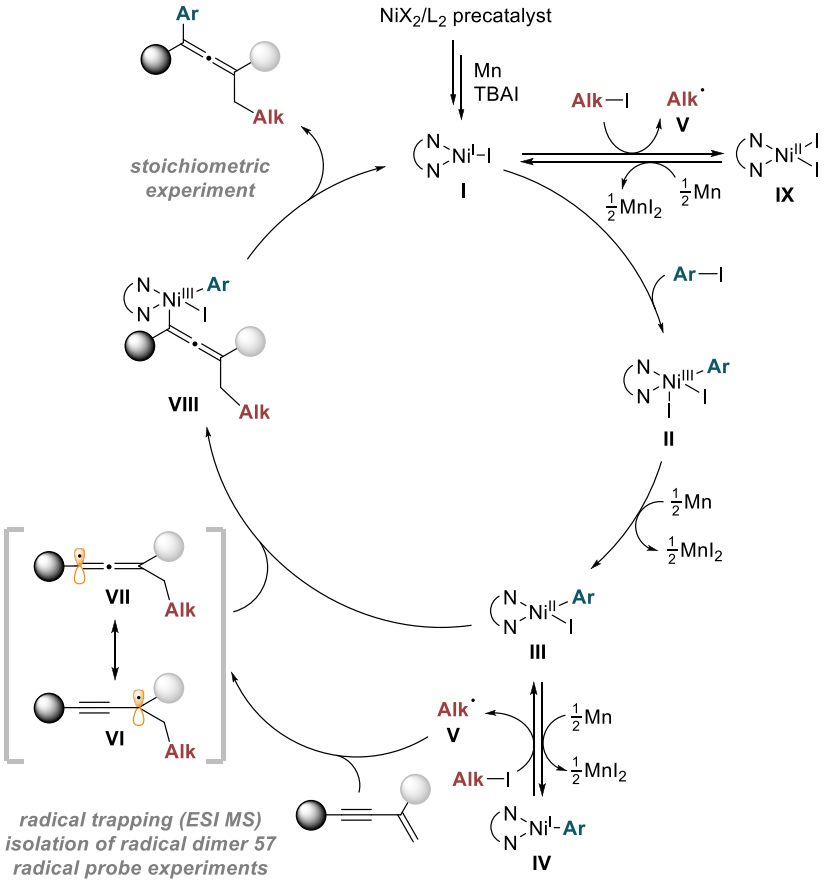
68, Ar = *p*-NCC_6H_4
69, Ar = Ph

4, 32%
not observed

57, 14%
57, 20%

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Scheme 7. Proposed Catalytic Cycle



formation of **57** from this reaction confirmed the presence of the propargyl/allenyl radical intermediate.

3. CONCLUSIONS

In summary, we have developed a nickel-catalyzed reductive 1,4-alkylarylation method for the catalytic conversion of 1,3-enynes into tri- or tetrasubstituted allenes. The selective transformation of conjugated π systems, including 1,3-enynes, has remained a challenging topic because of the presence of multiple active sites suitable for coupling reactions. Here, the conversion of 1,3-enynes into allenes was achieved using alkyl and aryl iodides and without the need for substrate preactivation or directing groups.

Our mechanistic investigation revealed a propargyl/allenyl radical as an intermediate in the catalytic reaction. We also investigated the reactivity of L_2NiAr complexes, which are expected to be involved in this reaction and in related Ni-catalyzed reductive difunctionalization reactions of unsaturated substrates. In particular, the complexes Ni(**L5**)(*p*-NCC_6H_4) and Ni(**L5**)(Ph) were generated in situ and characterized as Ni^I complexes. The reaction of Ni(**L5**)(*p*-NCC_6H_4) with alkyl iodide and 1,3-enyne formed the propargyl radical dimer and the allene product. This result indicates that this complex is a possible intermediate in the catalytic reaction described here and that it can promote the activation of alkyl iodide, the generation of the propargyl/allenyl radical, and the selective coupling of this radical with the aryl component. By contrast, the reaction of Ni(**L5**)(Ph), derived from the catalytically inactive phenyl iodide, formed only the propargyl radical dimer but not the allene. These results may suggest an electronic effect of the aryl component in the XEC reaction and offer a rationale for the scope of aryl iodides in the catalytic reaction. We anticipate that this work contributes to the expansion of the territory of conjunctive platforms for selectivity-controlled reductive three-component difunctionalization reactions.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.4c01189.

Detailed experimental procedures, characterization data, and spectroscopic data (PDF)

Crystallographic data for **25** (CIF)

Crystallographic data for **57** (CIF)

Accession Codes

CCDC 2150685 (for **25**) and 2150686 (for **57**) contain the supplementary crystallographic data for this paper. These data can be obtained free of charge from the Cambridge Crystallographic Data Centre.

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Notes

The authors declare no competing financial interest.

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